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Soft Robotics

Explosion Driven Soft Robotic Actuator

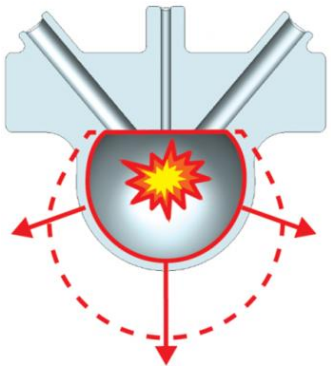
Introduction

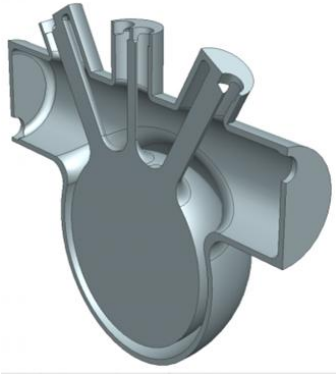
Soft robotics enables robots to interact safely and adaptively with complex environments.

Traditional actuation methods include pneumatics, hydraulics, and shape memory alloys.

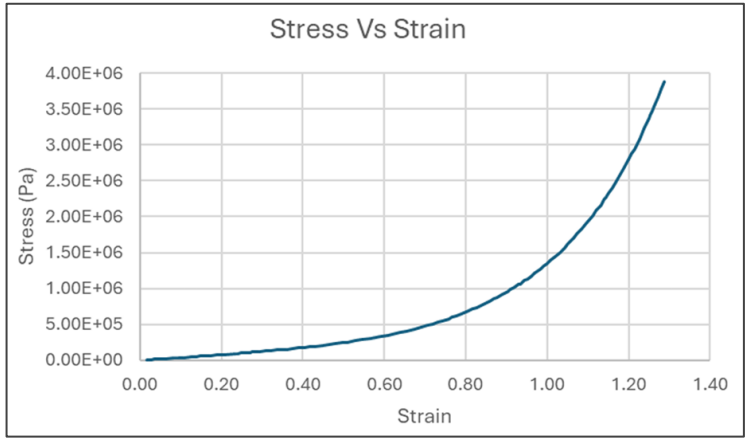
This project explores a novel method: *explosion-driven actuation*, using rapid, contained gas expansion to deform a soft elastomeric actuator.

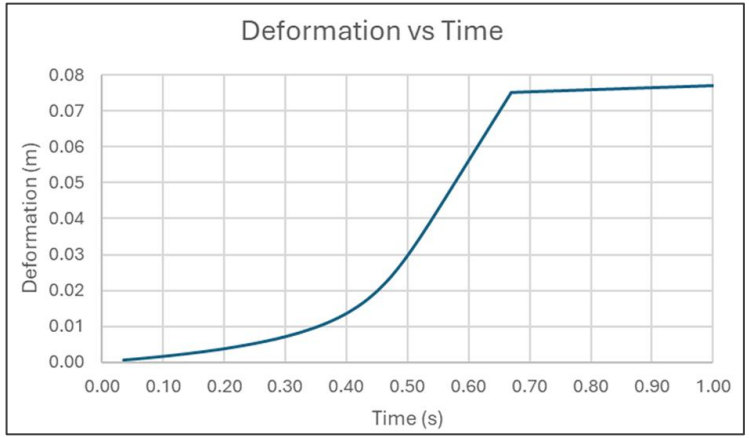
Objective: Model, simulate, and analyze the actuator's dynamic deformation, stress-strain response, and thermal behavior under explosive pressure.





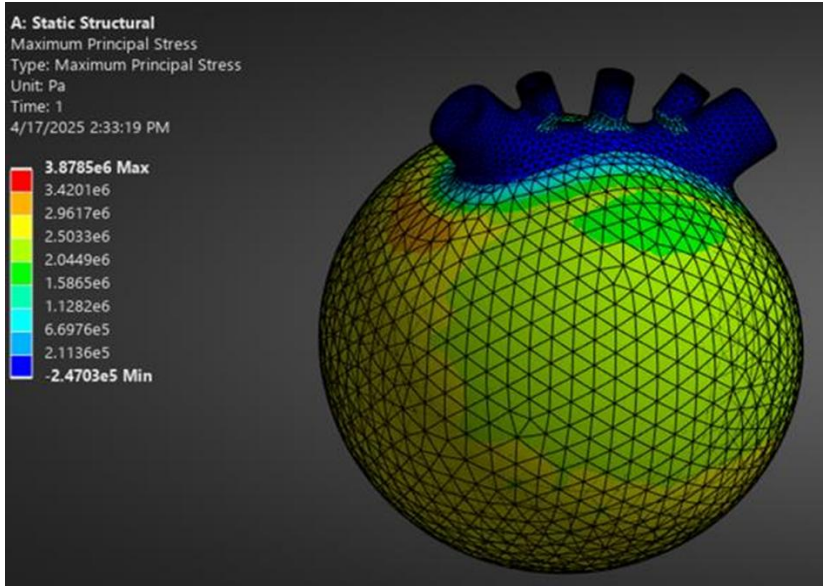
Analysis & Results

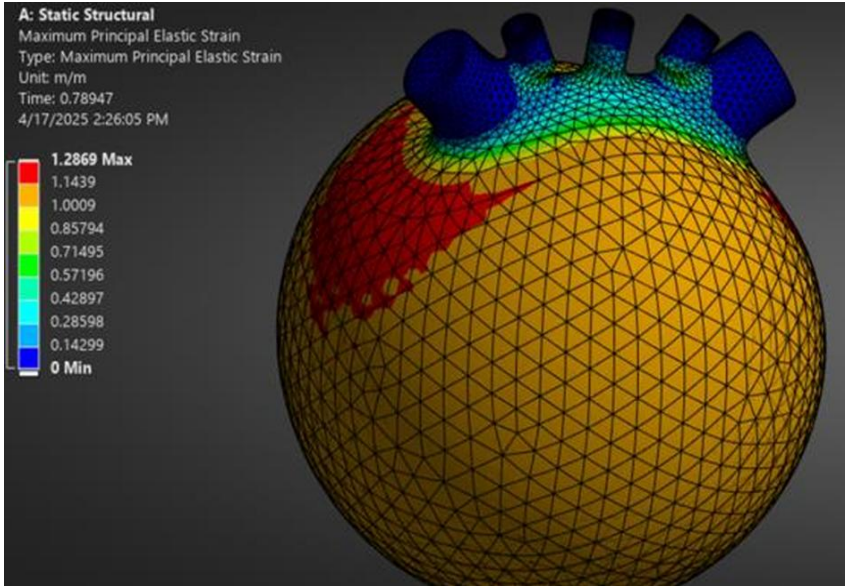




The actuator stayed within the elastic range of Ecoflex™ 00-30, showing no rupture at all during the simulated explosion-driven actuation.

Due to Ecoflex's hyperelastic nature, the stress-strain response was nonlinear yet monotonic, reflecting predictable deformation under the increasing pressure.





Principal Stress

The peak principal stress reached approximately **4 MPa.**

Principal Strain

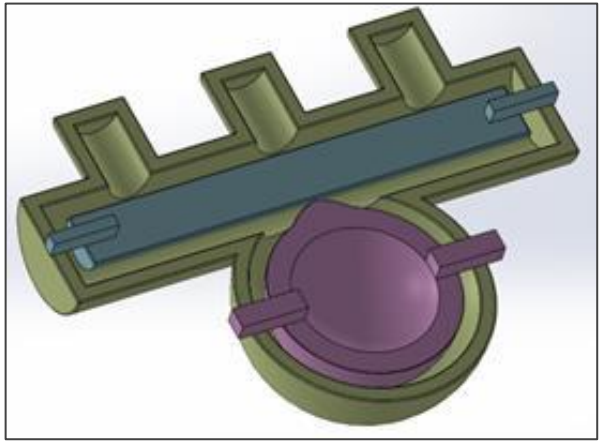
The max principal strain reaches to **0.07** during the pressure pulse.

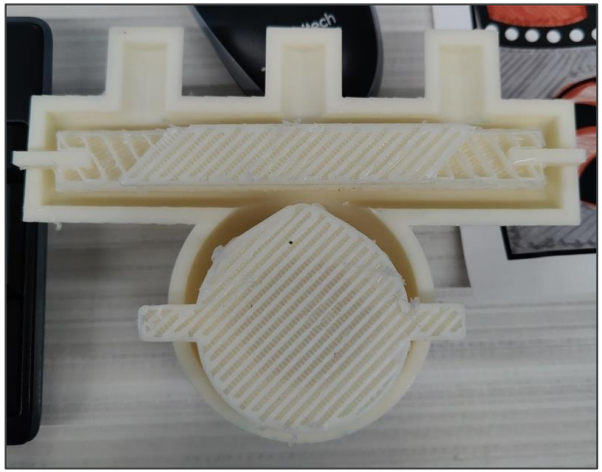
Designing & 3D Printing Molds

Using the 3D model of the part as reference and giving required tolerances for printing & pouring.

Incorporating resting stands in mold to allow uniform flow of material.

Creating the molds for 3D printing while maintaining the weight and dimension constraints.



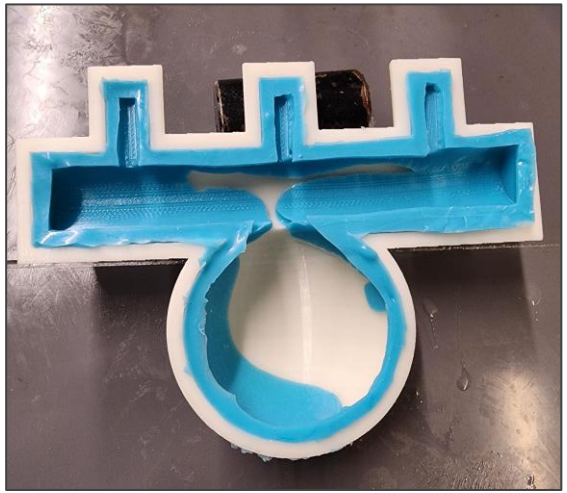


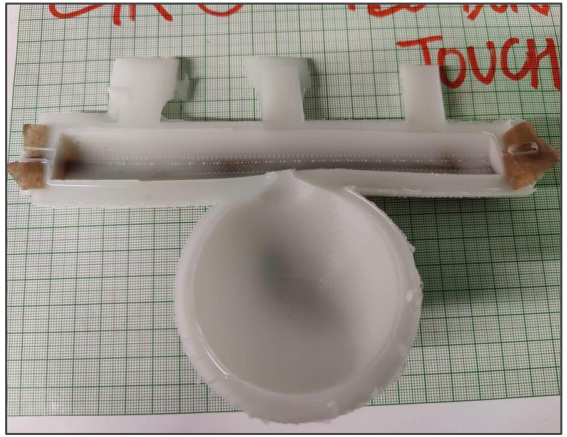
Mold Preparation & Pouring

Using Mold Star and then Ecoflex material for fabrication.

Mold Star was too viscous and stiff to properly flow into small gaps due to tight tolerances in the part.

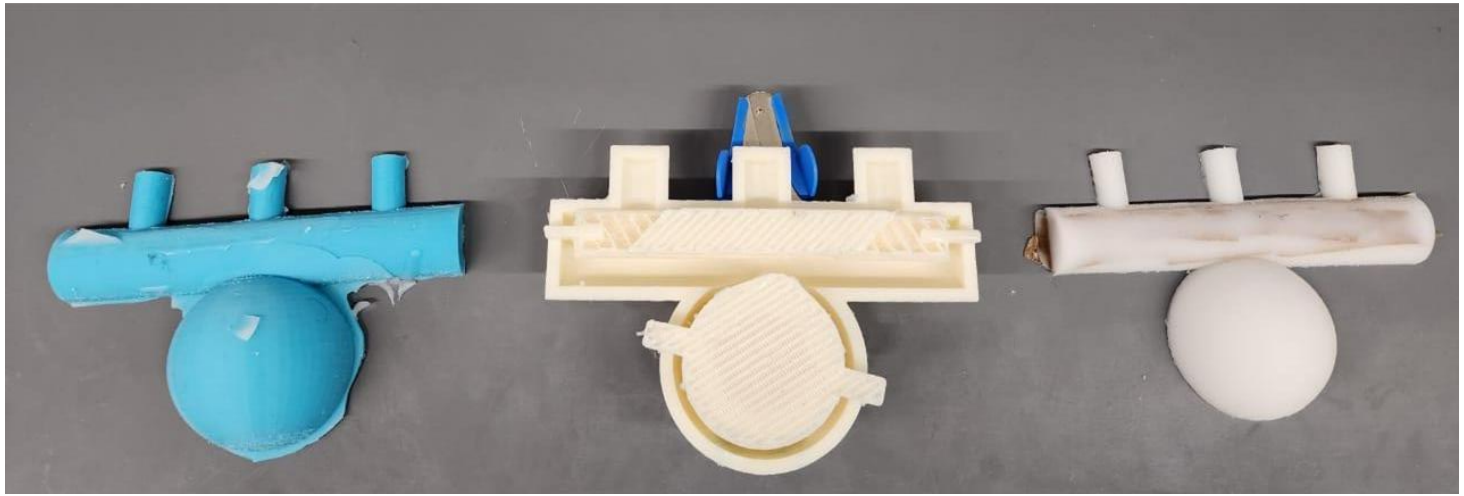
Ecoflex Material much suitable for the part.

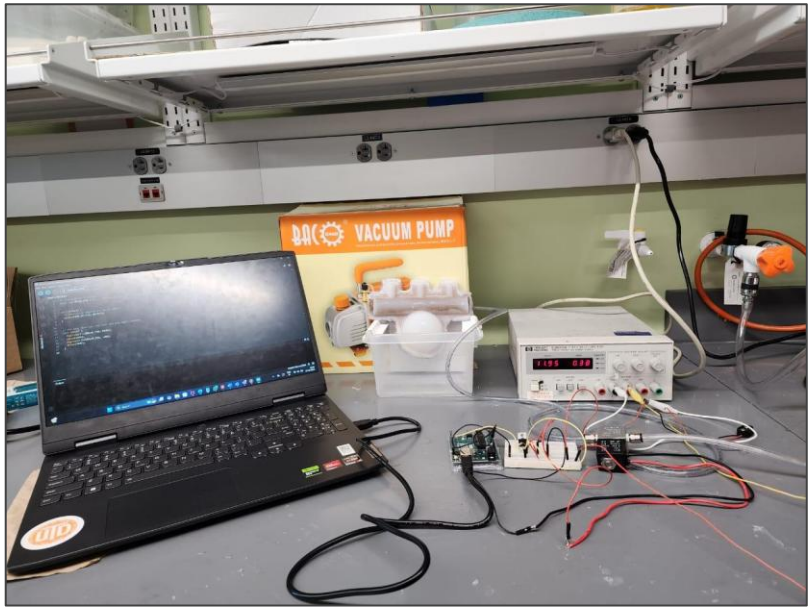




Material Comparison

After comparing the end result of all the different materials, **Ecoflex 010** came out to be the best fit. To provide variable stiffness using the paper strips to stiffen the top part of the robot.





Working

After supplying the compressed air from one end of the model, the chamber expands and thus providing actuation.

Connections

Using Arduino, creating the necessary connections and running the script to monitor the working under varying loads.

Conclusion

Explosion-driven actuation enables high-speed and large-amplitude motion in soft robots.

Ecoflex™ 00-30 actuator shows significant deformation under rapid internal pressurization while remaining thermally stable.

Suitable for rapid deployment, rescue devices, and jumping modules requiring short, intense bursts of motion.

Future work: Focus on experimental validation and design optimization for enhanced safety and efficiency.

Thank You